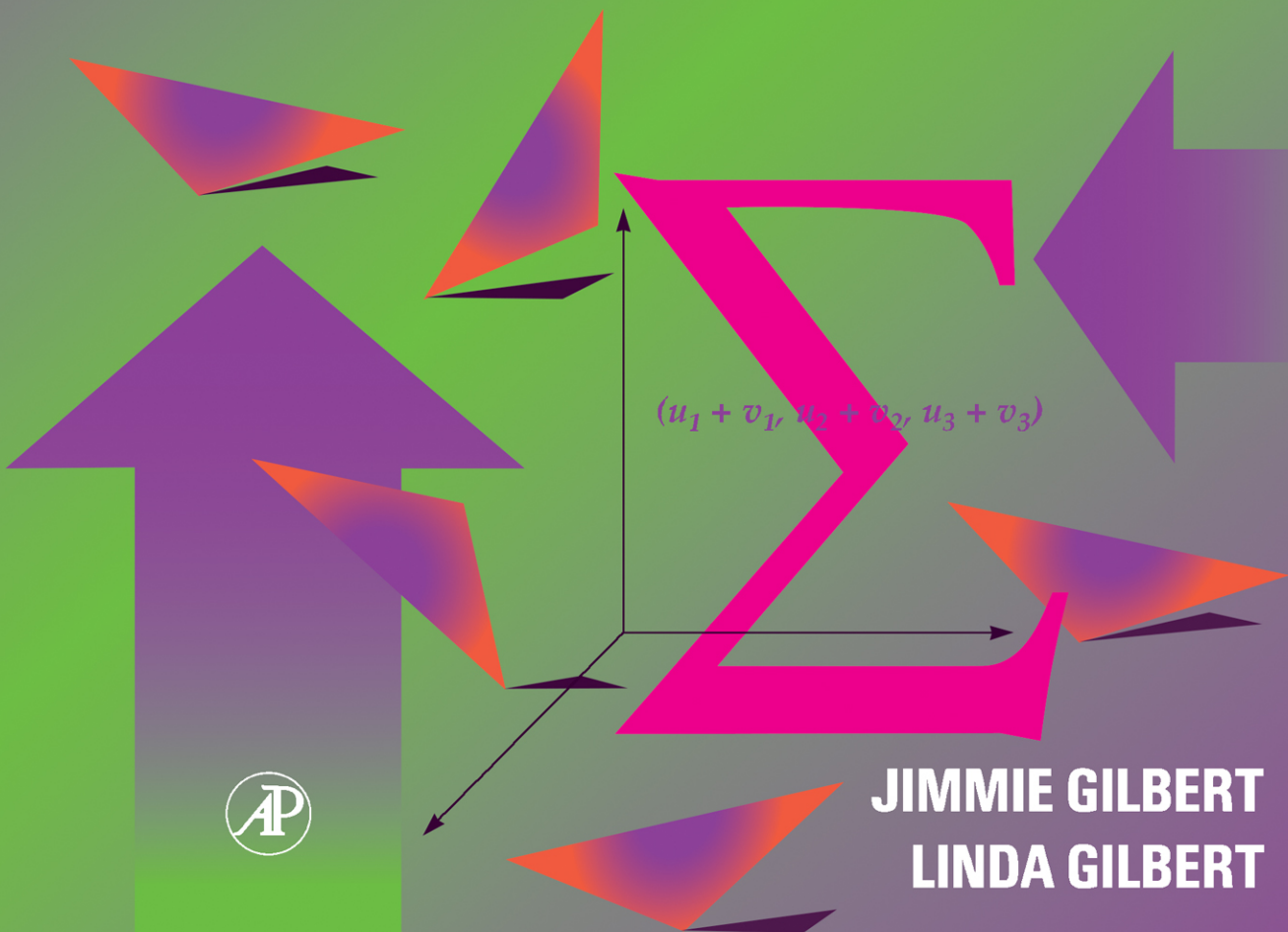


LINEAR ALGEBRA AND MATRIX THEORY



JIMMIE GILBERT
LINDA GILBERT

LINEAR ALGEBRA

AND MATRIX

THEORY

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AND MATRIX

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JIMMIE GILBERT

LINDA GILBERT

University of South Carolina at Spartanburg
Spartanburg, South Carolina



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Preface

This text was planned for use with one of the following two options:

- The complete text would be suitable for a one-year undergraduate course for mathematics majors and would provide a strong foundation for more abstract courses at higher levels of instruction.
- A one-semester or one-quarter course could be taught from the first five chapters together with selections from the remaining chapters. The selections could be chosen so as to meet the requirements of students in the fields of business, science, economics, or engineering.

The presentation of material presumes the knowledge and maturity gained from one calculus course, and a second calculus course is a desirable prerequisite.

It is our opinion that linear algebra is well suited to provide the transition from the intuitive developments of courses at a lower level to the more abstract treatments encountered later. Throughout the treatment here, material is presented from a structural point of view: fundamental algebraic properties of the entities involved are emphasized. This approach is particularly important because the mathematical systems encountered in linear algebra furnish a wealth of examples for the structures studied in more advanced courses.

The unifying concept for the first five chapters is that of elementary operations. This concept provides the pivot for a concise and efficient development of the basic theory of vector spaces, linear transformations, matrix multiplication, and the fundamental equivalence relations on matrices.

A rigorous treatment of determinants from the traditional viewpoint is presented in Chapter 6. For a class already familiar with this material, the chapter can be omitted.

In Chapters 7 through 10, the central theme of the development is the change in the matrix representing a vector function when only certain types of basis changes are admitted. It is from this approach that the classical canonical forms for matrices are derived.

Numerous examples and exercises are provided to illustrate the theory. Exercises are included of both computational and theoretical nature. Those of a theoretical nature amplify the treatment and provide experience in constructing deductive arguments, while those of a computational nature illustrate fundamental techniques. The amount of labor in the computational problems is kept to a minimum. Even so, many of them provide opportunities to utilize current technology, if that is the wish of the instructor. Answers are provided for about half of the computational problems.

The exercises are intended to develop confidence and deepen understanding. It is assumed that students grow in maturity as they progress through the text and the proportion of theoretical problems increases in later chapters.

Since much of the interest in linear algebra is due to its applications, the solution of systems of linear equations and the study of eigenvalue problems appear early in the text. Chapters 4 and 7 contain the most important applications of the theory.

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Jimmie Gilbert
Linda Gilbert

Chapter 1

Real Coordinate Spaces

1.1 Introduction

There are various approaches to that part of mathematics known as *linear algebra*. Different approaches emphasize different aspects of the subject such as matrices, applications, or computational methods. As presented in this text, linear algebra is in essence a study of vector spaces, and this study of vector spaces is primarily devoted to finite-dimensional vector spaces. The real coordinate spaces, in addition to being important in many applications, furnish excellent intuitive models of abstract finite-dimensional vector spaces. For these reasons, we begin our study of linear algebra with a study of the real coordinate spaces. Later it will be found that many of the results and techniques employed here will easily generalize to more abstract settings.

1.2 The Vector Spaces $\mathbf{R}^n$

Throughout this text the symbol $\mathbf{R}$ will denote the set of all real numbers. We assume a knowledge of the algebraic properties of $\mathbf{R}$, and begin with the following definition of real coordinate spaces.

Definition 1.1 For each positive integer n , $\mathbf{R}^n$ will denote the set of all ordered n -tuples $(u_1, u_2, \dots, u_n)$ of real numbers u_i . Two n -tuples $(u_1, u_2, \dots, u_n)$ and $(v_1, v_2, \dots, v_n)$ are equal if and only if $u_i = v_i$ for $i = 1, 2, \dots, n$. The set $\mathbf{R}^n$ is referred to as an **n -dimensional real coordinate space**. The elements of $\mathbf{R}^n$ are called **n -dimensional real coordinate vectors**, or simply **vectors**. The numbers u_i in a vector $(u_1, u_2, \dots, u_n)$ will be called the **components** of the vector. The elements of $\mathbf{R}$ will be referred to as **scalars**.¹

¹The terms “vector” and “scalar” are later extended to more general usage, but this will cause no confusion since the context will make the meaning clear.

The real coordinate spaces and the related terminology described in this definition are easily seen to be generalizations and extensions of the two- and three-dimensional vector spaces studied in the calculus.

When we use a single letter to represent a vector, the letter will be printed in boldface lower case Roman, such as $\mathbf{v}$, or written with an arrow over it, such as $\vec{v}$. In handwritten work with vectors, the arrow notation $\vec{v}$ is commonly used. Scalars will be represented by letters printed in lower case italics.

Definition 1.2 Addition in $\mathbf{R}^n$ is defined as follows: for any $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ in $\mathbf{R}^n$, the sum $\mathbf{u} + \mathbf{v}$ is given by

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n).$$

For any scalar a and any vector $\mathbf{u} = (u_1, u_2, \dots, u_n)$ in $\mathbf{R}^n$, the **product $a\mathbf{u}$** is defined by

$$a\mathbf{u} = (au_1, au_2, \dots, au_n).$$

The operation that combines the scalar a and the vector $\mathbf{u}$ to yield $a\mathbf{u}$ is referred to as **multiplication of the vector $\mathbf{u}$ by the scalar a** , or simply as **scalar multiplication**. Also, the product $a\mathbf{u}$ is called a **scalar multiple** of $\mathbf{u}$.

The following theorem gives the basic properties of the two operations that we have defined.

Theorem 1.3 *The following properties are valid for any scalars a and b , and any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbf{R}^n$:*

1. $\mathbf{u} + \mathbf{v} \in \mathbf{R}^n$. (Closure under addition)
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$. (Associative property of addition)
3. There is a vector $\mathbf{0}$ in $\mathbf{R}^n$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$ for all $\mathbf{u} \in \mathbf{R}^n$. (Additive identity)
4. For each $\mathbf{u} \in \mathbf{R}^n$, there is a vector $-\mathbf{u}$ in $\mathbf{R}^n$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$. (Additive inverses)
5. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$. (Commutative property of addition)
6. $a\mathbf{u} \in \mathbf{R}^n$. (Absorption under scalar multiplication)
7. $a(b\mathbf{u}) = (ab)\mathbf{u}$. (Associative property of scalar multiplication)
8. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$. (Distributive property, vector addition)
9. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$. (Distributive property, scalar addition)
10. $1 \cdot \mathbf{u} = \mathbf{u}$.

The proofs of these properties are easily carried out using the definitions of vector addition and scalar multiplication, as well as the properties of real numbers. As typical examples, properties 3, 4, and 8 will be proved here. The remaining proofs are left as an exercise.

Proof of Property 3. The vector $\mathbf{0} = (0, 0, \dots, 0)$ is in $\mathbf{R}^n$, and if $\mathbf{u} = (u_1, u_2, \dots, u_n)$,

$$\mathbf{u} + \mathbf{0} = (u_1 + 0, u_2 + 0, \dots, u_n + 0) = \mathbf{u}.$$

Proof of Property 4. If $\mathbf{u} = (u_1, u_2, \dots, u_n) \in \mathbf{R}^n$, then $\mathbf{v} = (-u_1, -u_2, \dots, -u_n)$ is in $\mathbf{R}^n$ since all real numbers have additive inverses. And since

$$\mathbf{u} + \mathbf{v} = (u_1 + (-u_1), u_2 + (-u_2), \dots, u_n + (-u_n)) = \mathbf{0},$$

$\mathbf{v}$ is the vector $-\mathbf{u}$ as required by property (4).

Proof of Property 8. Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$. Then

$$\begin{aligned} a(\mathbf{u} + \mathbf{v}) &= a(u_1 + v_1, u_2 + v_2, \dots, u_n + v_n) \\ &= (a(u_1 + v_1), a(u_2 + v_2), \dots, a(u_n + v_n)) \\ &= (au_1 + av_1, au_2 + av_2, \dots, au_n + av_n) \\ &= (au_1, au_2, \dots, au_n) + (av_1, av_2, \dots, av_n) \\ &= a\mathbf{u} + a\mathbf{v}. \end{aligned} \quad \blacksquare$$

The associative property of addition in $\mathbf{R}^n$ can be generalized so that the terms in a sum such as $a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + \dots + a_k\mathbf{u}_k$ can be grouped with parentheses in any way without changing the value of the result. Such a sum can be written in the compact form $\sum_{i=1}^k a_i\mathbf{u}_i$ or $\sum_i a_i\mathbf{u}_i$ if the number of terms in the sum is not important. It is always understood that only a finite number of nonzero terms is involved in such a sum.

Definition 1.4 Let $\mathcal{A}$ be a nonempty set of vectors in $\mathbf{R}^n$. A vector $\mathbf{v}$ in $\mathbf{R}^n$ is **linearly dependent** on the set $\mathcal{A}$ if there exist vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ in $\mathcal{A}$ and scalars $a_1, a_2, \dots, a_k$ such that $\mathbf{v} = \sum_{i=1}^k a_i\mathbf{u}_i$. A vector of the form $\sum_{i=1}^k a_i\mathbf{u}_i$ is called a **linear combination** of the $\mathbf{u}_i$.

If linear dependence on a finite set $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ is under consideration, the statement in Definition 1.4 is equivalent to requiring that *all* of the vectors in $\mathcal{B}$ be involved in the linear combination. That is, a vector $\mathbf{v}$ is linearly dependent on $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ if and only if there are scalars $b_1, b_2, \dots, b_r$ such that $\mathbf{v} = \sum_{i=1}^r b_i\mathbf{v}_i$.

Example 1 $\square$ As an illustration, let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = (1, 0, 2, 1), \mathbf{v}_2 = (0, -2, 2, 3), \text{ and } \mathbf{v}_3 = (1, -2, 4, 4),$$

and let $\mathbf{v} = (3, -4, 10, 9)$. Now $\mathbf{v}$ can be written as

$$\mathbf{v} = 1 \cdot \mathbf{v}_1 + 0 \cdot \mathbf{v}_2 + 2 \cdot \mathbf{v}_3,$$

or

$$\mathbf{v} = 3 \cdot \mathbf{v}_1 + 2 \cdot \mathbf{v}_2 + 0 \cdot \mathbf{v}_3.$$

Either of these combinations shows that $\mathbf{v}$ is linearly dependent on $\mathcal{B}$.

To consider a situation involving an infinite set, let $\mathcal{A}$ be the set of all vectors in $\mathbf{R}^3$ that have integral components, and let $\mathbf{v} = (\sqrt{2}, \frac{2}{3}, 0)$. Now $\mathbf{u}_1 = (1, 0, 1)$, $\mathbf{u}_2 = (0, 1, 0)$, and $\mathbf{u}_3 = (0, 0, 1)$ are in $\mathcal{A}$, and

$$\mathbf{v} = \sqrt{2}\mathbf{u}_1 + \frac{2}{3}\mathbf{u}_2 - \sqrt{2}\mathbf{u}_3.$$

Thus $\mathbf{v}$ is linearly dependent on $\mathcal{A}$. It should be noted that other choices of vectors $\mathbf{u}_i$ can be made in order to exhibit this dependence. ■

In order to decide whether a certain vector is linearly dependent on a given set in $\mathbf{R}^n$, it is usually necessary to solve a system of equations. This is illustrated in the following example.

Example 2 □ Consider the question as to whether $(6, 0, -1)$ is linearly dependent on the set $\mathcal{A} = \{(2, -1, 1), (0, 1, -1), (-2, 1, 0)\}$. To answer the question, we investigate the conditions on a_1, a_2 , and a_3 that are required by the equation

$$a_1(2, -1, 1) + a_2(0, 1, -1) + a_3(-2, 1, 0) = (6, 0, -1).$$

Performing the indicated scalar multiplications and additions in the left member of this equation leads to

$$(2a_1 - 2a_3, -a_1 + a_2 + a_3, a_1 - a_2) = (6, 0, -1).$$

This vector equation is equivalent to the system of equations

$$\begin{aligned} 2a_1 & & - 2a_3 & = & 6 \\ -a_1 + a_2 + a_3 & = & 0 \\ a_1 - a_2 & & & = & -1. \end{aligned}$$

We decide to work toward the solution of this system by eliminating a_1 from two of the equations in the system. As steps toward this goal, we multiply the first equation by $\frac{1}{2}$ and we add the second equation to the third equation. These steps yield the system

$$\begin{aligned} a_1 & & - a_3 & = & 3 \\ -a_1 + a_2 + a_3 & = & 0 \\ a_3 & = & -1. \end{aligned}$$

Adding the first equation to the second now results in

$$\begin{aligned} a_1 & & - a_3 & = & 3 \\ a_2 & & & = & 3 \\ a_3 & = & -1. \end{aligned}$$

The solution $a_1 = 2, a_2 = 3, a_3 = -1$ is now readily obtained. Thus the vector $(6, 0, -1)$ is linearly dependent on the set $\mathcal{A}$. ■

Another important type of dependence is given in the definition below. This time, the phrase *linearly dependent* involves only a set instead of involving both a vector and a set.

Definition 1.5 A set $\mathcal{A}$ of vectors in $\mathbf{R}^n$ is **linearly dependent** if there is a collection of vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ in $\mathcal{A}$ and scalars $c_1, c_2, \dots, c_k$, not all of which are zero, such that $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k = \mathbf{0}$. If no such collection of vectors exists in $\mathcal{A}$, then $\mathcal{A}$ is called **linearly independent**.

Again, the case involving a finite set of vectors is of special interest. It is readily seen that a finite set $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ is linearly dependent if and only if there are scalars $b_1, b_2, \dots, b_r$, not all zero, such that $\sum_{i=1}^r b_i \mathbf{v}_i = \mathbf{0}$.

Example 3 $\square$ The set $\mathcal{B}$ in Example 1 is linearly dependent since $\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$. The set $\mathcal{A}$ is also linearly dependent. For if $\mathbf{u}_1, \mathbf{u}_2$, and $\mathbf{u}_3$ are as before and $\mathbf{u}_4 = (-3, -2, -4)$, then $3\mathbf{u}_1 + 2\mathbf{u}_2 + \mathbf{u}_3 + \mathbf{u}_4 = \mathbf{0}$. $\blacksquare$

To determine the linear dependence or linear independence of a set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ of vectors in $\mathbf{R}^n$, it is necessary to investigate the conditions on the c_i which are imposed by requiring that $\sum_{j=1}^k c_j \mathbf{u}_j = \mathbf{0}$. If $\mathbf{u}_j = (u_{1j}, u_{2j}, \dots, u_{nj})$, we have

$$\begin{aligned} \sum_{j=1}^k c_j \mathbf{u}_j &= \sum_{j=1}^k c_j (u_{1j}, u_{2j}, \dots, u_{nj}) \\ &= \left(\sum_{j=1}^k c_j u_{1j}, \sum_{j=1}^k c_j u_{2j}, \dots, \sum_{j=1}^k c_j u_{nj} \right). \end{aligned}$$

Thus $\sum_{j=1}^k c_j \mathbf{u}_j = \mathbf{0}$ if and only if $\sum_{j=1}^k c_j u_{ij} = 0$ for each $i = 1, 2, \dots, n$. This shows that the problem of determining the conditions on the c_j is equivalent to investigating the solutions of a system of n equations in k unknowns. If $\sum_{j=1}^k c_j \mathbf{u}_j = \mathbf{0}$ implies $c_1 = c_2 = \dots = c_k = 0$, then $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is linearly independent.

The discussion in the preceding paragraph is illustrated in the next example.

Example 4 $\square$ Consider the set of vectors

$$\{(1, 1, 8, 1), (1, 0, 3, 0), (3, 1, 14, 1)\}$$

in $\mathbf{R}^4$. To determine the linear dependence or linear independence of this set, we investigate the solutions c_1, c_2, c_3 to

$$c_1(1, 1, 8, 1) + c_2(1, 0, 3, 0) + c_3(3, 1, 14, 1) = (0, 0, 0, 0).$$

Performing the indicated vector operations gives the equation

$$(c_1 + c_2 + 3c_3, c_1 + c_3, 8c_1 + 3c_2 + 14c_3, c_1 + c_3) = (0, 0, 0, 0).$$

This vector equation is equivalent to

$$\begin{aligned} c_1 + c_2 + 3c_3 &= 0 \\ c_1 + c_3 &= 0 \\ 8c_1 + 3c_2 + 14c_3 &= 0 \\ c_1 + c_3 &= 0. \end{aligned}$$

To solve this system, we first interchange the first two equations to place the equation $c_1 + c_3 = 0$ at the top.

$$\begin{aligned} c_1 + c_3 &= 0 \\ c_1 + c_2 + 3c_3 &= 0 \\ 8c_1 + 3c_2 + 14c_3 &= 0 \\ c_1 + c_3 &= 0 \end{aligned}$$

By adding suitable multiples of the first equation to each of the other equations, we then eliminate c_1 from all but the first equation. This yields the system

$$\begin{aligned} c_1 + c_3 &= 0 \\ c_2 + 2c_3 &= 0 \\ 3c_2 + 6c_3 &= 0 \\ 0 &= 0. \end{aligned}$$

Eliminating c_2 from the third equation gives

$$\begin{aligned} c_1 + c_3 &= 0 \\ c_2 + 2c_3 &= 0 \\ 0 &= 0 \\ 0 &= 0. \end{aligned}$$

It is now clear that there are many solutions to the system, and they are given by

$$\begin{aligned} c_1 &= -c_3 \\ c_2 &= -2c_3 \\ c_3 &\text{ is arbitrary.} \end{aligned}$$

In particular, it is not necessary that c_3 be zero, so the original set of vectors is linearly dependent. ■

The following theorem gives an alternative description of linear dependence for a certain type of set.

Theorem 1.6 *A set $\mathcal{A} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ in $\mathbf{R}^n$ that contains at least two vectors is linearly dependent if and only if some $\mathbf{u}_j$ is a linear combination of the remaining vectors in the set.*

Proof. If the set $\mathcal{A} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is linearly dependent, then there are scalars $a_1, a_2, \dots, a_k$ such that $\sum_{i=1}^k a_i \mathbf{u}_i = \mathbf{0}$ with at least one a_i , say a_j , not zero. This implies that

$$a_j \mathbf{u}_j = -a_1 \mathbf{u}_1 - \cdots - a_{j-1} \mathbf{u}_{j-1} - a_{j+1} \mathbf{u}_{j+1} - \cdots - a_k \mathbf{u}_k$$

so that

$$\mathbf{u}_j = \left(-\frac{a_1}{a_j}\right) \mathbf{u}_1 + \cdots + \left(-\frac{a_{j-1}}{a_j}\right) \mathbf{u}_{j-1} + \left(-\frac{a_{j+1}}{a_j}\right) \mathbf{u}_{j+1} + \cdots + \left(-\frac{a_k}{a_j}\right) \mathbf{u}_k.$$

Thus $\mathbf{u}_j$ can be written as a linear combination of the remaining vectors in the set.

Now assume that some $\mathbf{u}_j$ is a linear combination of the remaining vectors in the set, i.e.,

$$\mathbf{u}_j = b_1 \mathbf{u}_1 + b_2 \mathbf{u}_2 + \cdots + b_{j-1} \mathbf{u}_{j-1} + b_{j+1} \mathbf{u}_{j+1} + \cdots + b_k \mathbf{u}_k.$$

Then

$$b_1 \mathbf{u}_1 + \cdots + b_{j-1} \mathbf{u}_{j-1} + (-1) \mathbf{u}_j + b_{j+1} \mathbf{u}_{j+1} + \cdots + b_k \mathbf{u}_k = \mathbf{0},$$

and since the coefficient of $\mathbf{u}_j$ in this linear combination is not zero, the set is linearly dependent. ■

The different meanings of the word “dependent” in Definitions 1.4 and 1.5 should be noted carefully. These meanings, though different, are closely related. The preceding theorem, for example, could be restated as follows: “A set $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is linearly dependent if and only if some $\mathbf{u}_k$ is linearly dependent on the remaining vectors.” This relation is further illustrated in some of the exercises at the end of this section.

In the last section of this chapter, the following definition and theorem are of primary importance. Both are natural extensions of Definition 1.4.

Definition 1.7 *A set $\mathcal{A} \subseteq \mathbf{R}^n$ is linearly dependent on the set $\mathcal{B} \subseteq \mathbf{R}^n$ if each $\mathbf{u} \in \mathcal{A}$ is linearly dependent on $\mathcal{B}$.*

Thus $\mathcal{A}$ is linearly dependent on $\mathcal{B}$ if and only if each vector in $\mathcal{A}$ is a linear combination of vectors that are contained in $\mathcal{B}$.

Other types of dependence are considered in some situations, but linear dependence is the only type that we will use. For this reason, we will frequently use the term “dependent” in place of “linearly dependent.”

In the proof of the next theorem, it is convenient to have available the notation $\sum_{i=1}^k \left(\sum_{j=1}^m \mathbf{u}_{ij} \right)$ for the sum of the form $\sum_{j=1}^m \mathbf{u}_{1j} + \sum_{j=1}^m \mathbf{u}_{2j} + \cdots + \sum_{j=1}^m \mathbf{u}_{kj}$. The

associative and commutative properties for vector addition [Theorem 1.3, (2) and (5)] imply that

$$\sum_{i=1}^k \left(\sum_{j=1}^m \mathbf{u}_{ij} \right) = \sum_{j=1}^m \left(\sum_{i=1}^k \mathbf{u}_{ij} \right).$$

Theorem 1.8 *Let $\mathcal{A}, \mathcal{B}$, and $\mathcal{C}$ be subsets of $\mathbf{R}^n$. If $\mathcal{A}$ is dependent on $\mathcal{B}$ and $\mathcal{B}$ is dependent on $\mathcal{C}$, then $\mathcal{A}$ is dependent on $\mathcal{C}$.*

Proof. Suppose that $\mathcal{A}$ is dependent on $\mathcal{B}$ and $\mathcal{B}$ is dependent on $\mathcal{C}$, and let $\mathbf{u}$ be an arbitrary vector in $\mathcal{A}$. Since $\mathcal{A}$ is dependent on $\mathcal{B}$, there are vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ in $\mathcal{B}$ and scalars $a_1, a_2, \dots, a_m$ such that

$$\mathbf{u} = \sum_{j=1}^m a_j \mathbf{v}_j.$$

Since $\mathcal{B}$ is dependent on $\mathcal{C}$, each $\mathbf{v}_j$ can be written as a linear combination of certain vectors in $\mathcal{C}$. In general, for different vectors $\mathbf{v}_j$, different sets of vectors from $\mathcal{C}$ would be involved in these linear combinations. But each of these m linear combinations (one for each $\mathbf{v}_j$) would involve only a finite number of vectors. Hence the set of all vectors in $\mathcal{C}$ that appear in a term of at least one of these linear combinations is a finite set, say $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$, and each $\mathbf{v}_j$ can be written in the form $\mathbf{v}_j = \sum_{i=1}^k b_{ij} \mathbf{w}_i$. Replacing the $\mathbf{v}_j$'s in the above expression for $\mathbf{u}$ by these linear combinations, we obtain

$$\begin{aligned} \mathbf{u} &= \sum_{j=1}^m a_j \mathbf{v}_j \\ &= \sum_{j=1}^m a_j \left(\sum_{i=1}^k b_{ij} \mathbf{w}_i \right) \\ &= \sum_{j=1}^m \left(\sum_{i=1}^k a_j b_{ij} \mathbf{w}_i \right) \\ &= \sum_{i=1}^k \left(\sum_{j=1}^m a_j b_{ij} \mathbf{w}_i \right) \\ &= \sum_{i=1}^k \left(\sum_{j=1}^m a_j b_{ij} \right) \mathbf{w}_i. \end{aligned}$$

Letting $c_i = \sum_{j=1}^m a_j b_{ij}$ for $i = 1, 2, \dots, k$, we have $\mathbf{u} = \sum_{i=1}^k c_i \mathbf{w}_i$. Thus $\mathbf{u}$ is dependent on $\mathcal{C}$.

Since $\mathbf{u}$ was arbitrarily chosen in $\mathcal{A}$, $\mathcal{A}$ is dependent on $\mathcal{C}$ and the theorem is proved. ■

Exercises 1.2

1. For any pair of positive integers i and j , the symbol δ_{ij} is defined by $\delta_{ij} = 0$ if $i \neq j$ and $\delta_{ij} = 1$ if $i = j$. This symbol is known as the **Kronecker delta**.

- (a) Find the value of $\sum_{i=1}^n \left(\sum_{j=1}^m \delta_{ij} \right)$.
- (b) Find the value of $\sum_{i=1}^n \left(\sum_{j=1}^n (1 - \delta_{ij}) \right)$.
- (c) Find the value of $\sum_{i=1}^5 \left(\sum_{j=1}^4 (-1)^{\delta_{ij}} \right)$.
- (d) Find the value of $\sum_{j=1}^n \delta_{ij} \delta_{jk}$.

2. Prove the remaining parts of Theorem 1.3.

3. In each case, determine whether or not the given vector $\mathbf{v}$ is linearly dependent on the given set $\mathcal{A}$.

- (a) $\mathbf{v} = (-2, 1, 4)$, $\mathcal{A} = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$
- (b) $\mathbf{v} = (-4, 4, 2)$, $\mathcal{A} = \{(2, 1, -3), (1, -1, 3)\}$
- (c) $\mathbf{v} = (1, 2, 1)$, $\mathcal{A} = \{(1, 0, -2), (0, 2, 1), (1, 2, -1), (-1, 2, 3)\}$
- (d) $\mathbf{v} = (2, 13, -5)$, $\mathcal{A} = \{(1, 2, -1), (3, 6, -3), (-1, 1, 0), (0, 6, -2), (2, 4, -2)\}$
- (e) $\mathbf{v} = (0, 1, 2, 0)$, $\mathcal{A} = \{(1, 0, -1, 1), (-2, 0, 2, -2), (1, 1, 1, 1), (2, 1, 0, 2)\}$
- (f) $\mathbf{v} = (2, 3, 5, 5)$, $\mathcal{A} = \{(0, 1, 1, 1), (1, 0, 1, 1), (1, -1, 0, 0), (1, 1, 2, 2)\}$

4. Assuming the properties stated in Theorem 1.3, prove the following statements.

- (a) The vector $\mathbf{0}$ in property (3) is unique.
- (b) The vector $-\mathbf{u}$ in $\mathbf{R}^n$ which satisfies $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ is unique.
- (c) $-\mathbf{u} = (-1)\mathbf{u}$.
- (d) The vector $\mathbf{u} - \mathbf{v}$ is by definition the vector $\mathbf{w}$ such that $\mathbf{v} + \mathbf{w} = \mathbf{u}$. Prove that $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-1)\mathbf{v}$.

5. Determine whether or not the given set $\mathcal{A}$ is linearly dependent.

- (a) $\mathcal{A} = \{(1, 0, -2), (0, 2, 1), (-1, 2, 3)\}$
- (b) $\mathcal{A} = \{(1, 4, 3), (2, 12, 6), (5, 21, 15), (0, 2, -1)\}$
- (c) $\mathcal{A} = \{(1, 2, -1), (-1, 1, 0), (1, 3, -1)\}$
- (d) $\mathcal{A} = \{(1, 0, 1, 2), (2, 1, 0, 0), (4, 5, 6, 0), (1, 1, 1, 0)\}$

6. Show that the given set is linearly dependent and write one of the vectors as a linear combination of the remaining vectors.

- (a) $\{(2, 1, 0), (1, 1, 0), (0, 1, 1), (-1, 1, 1)\}$
- (b) $\{(1, 2, 1, 0), (3, -4, 5, 6), (2, -1, 3, 3), (-2, 6, -4, -6)\}$

7. Show that any vector in $\mathbf{R}^3$ is dependent on the set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ where $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$, $\mathbf{e}_3 = (0, 0, 1)$.

8. Show that every vector in $\mathbf{R}^3$ is dependent on the set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ where $\mathbf{u}_1 = (1, 0, 0)$, $\mathbf{u}_2 = (1, 1, 0)$, and $\mathbf{u}_3 = (1, 1, 1)$.
9. Show that there is one vector in the set

$$\{(1, 1, 0), (0, 1, 1), (1, 0, -1), (1, 0, 1)\}$$

that cannot be written as a linear combination of the other vectors in the set.

10. Prove that if the set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ of vectors in $\mathbf{R}^n$ contains the zero vector, it is linearly dependent.
11. Prove that a set consisting of exactly one nonzero vector is linearly independent.
12. Prove that a set of two vectors in $\mathbf{R}^n$ is linearly dependent if and only if one of the vectors is a scalar multiple of the other.
13. Prove that a set of nonzero vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ in $\mathbf{R}^n$ is linearly dependent if and only if some $\mathbf{u}_r$ is a linear combination of the preceding vectors.
14. Let $\mathcal{A} = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a linearly independent set of vectors in $\mathbf{R}^n$.
- Prove that the set $\{\mathbf{u}_1 - \mathbf{u}_2, \mathbf{u}_2 - \mathbf{u}_3, \mathbf{u}_1 + \mathbf{u}_3\}$ is linearly independent.
 - Prove that the set $\{\mathbf{u}_1 - \mathbf{u}_2, \mathbf{u}_2 - \mathbf{u}_3, \mathbf{u}_1 - \mathbf{u}_3\}$ is linearly dependent.
15. Let $\{\mathbf{u}_1, \dots, \mathbf{u}_{r-1}, \mathbf{u}_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_k\}$ be a linearly independent set of k vectors in $\mathbf{R}^n$, and let $\mathbf{u}'_r = \sum_{j=1}^k a_j \mathbf{u}_j$ with $a_r \neq 0$. Prove that $\{\mathbf{u}_1, \dots, \mathbf{u}_{r-1}, \mathbf{u}'_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_k\}$ is linearly independent.
16. Let $\emptyset$ denote the empty set of vectors in $\mathbf{R}^n$. Determine whether or not $\emptyset$ is linearly dependent, and justify your conclusion.
17. Prove that any subset of a linearly independent set $\mathcal{A} \subseteq \mathbf{R}^n$ is linearly independent.
18. Let $\mathcal{A} \subseteq \mathbf{R}^n$. Prove that if $\mathcal{A}$ contains a linearly dependent subset, then $\mathcal{A}$ is linearly dependent.

1.3 Subspaces of $\mathbf{R}^n$

There are many subsets of $\mathbf{R}^n$ that possess the properties stated in Theorem 1.3. A study of these subsets furnishes a great deal of insight into the structure of the spaces $\mathbf{R}^n$, and is of vital importance in subsequent material.

Definition 1.9 A set $\mathbf{W}$ is a **subspace** of $\mathbf{R}^n$ if $\mathbf{W}$ is contained in $\mathbf{R}^n$ and if the properties of Theorem 1.3 are valid in $\mathbf{W}$. That is,

- $\mathbf{u} + \mathbf{v} \in \mathbf{W}$ for all $\mathbf{u}, \mathbf{v}$ in $\mathbf{W}$.